



Graphs in Machine Learning

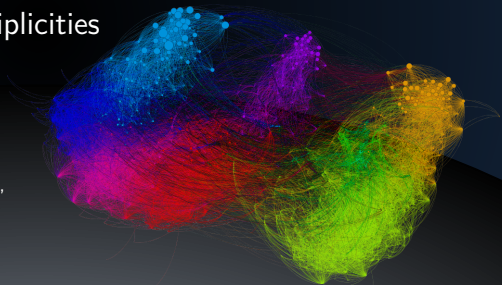
Online SSL: Graph Quantization

Charikar's Algorithm and Multiplicities

Michal Valko

Isara Labs

Partially based on material by: Branislav Kveton,
Mikhail Belkin, Jerry Zhu



Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

Keeps up to k centers $C_t = \{\mathbf{c}_1, \mathbf{c}_2, \dots\}$ with

Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

Keeps up to k centers $C_t = \{c_1, c_2, \dots\}$ with

- Distance $c_i, c_j \in C_t$ is at least $\geq R$

Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

Keeps up to k centers $C_t = \{\mathbf{c}_1, \mathbf{c}_2, \dots\}$ with

- Distance $\mathbf{c}_i, \mathbf{c}_j \in C_t$ is at least $\geq R$
- For each new $\mathbf{x}_t$, distance to some $\mathbf{c}_i \in C_t$ is less than R .

Online SSL with Graphs: Graph Quantization

An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

Keeps up to k centers $C_t = \{\mathbf{c}_1, \mathbf{c}_2, \dots\}$ with

- Distance $\mathbf{c}_i, \mathbf{c}_j \in C_t$ is at least $\geq R$
- For each new $\mathbf{x}_t$, distance to some $\mathbf{c}_i \in C_t$ is less than R .
- $|C_t| \leq k$

Online SSL with Graphs: Graph Quantization

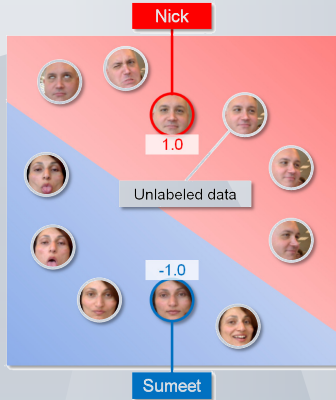
An idea: incremental k -centers

Doubling algorithm of Charikar et al. **charikar1997incremental**

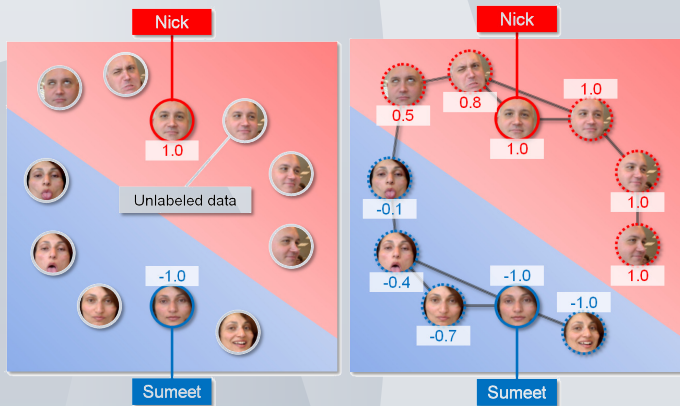
Keeps up to k centers $C_t = \{\mathbf{c}_1, \mathbf{c}_2, \dots\}$ with

- Distance $\mathbf{c}_i, \mathbf{c}_j \in C_t$ is at least $\geq R$
- For each new $\mathbf{x}_t$, distance to some $\mathbf{c}_i \in C_t$ is less than R .
- $|C_t| \leq k$
- if not possible, R is doubled

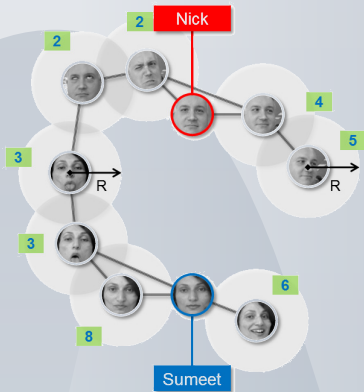
Online SSL with Graphs: Graph Quantization



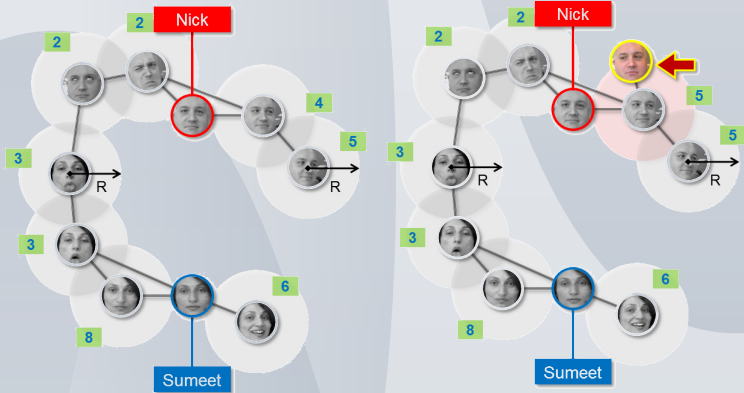
Online SSL with Graphs: Graph Quantization



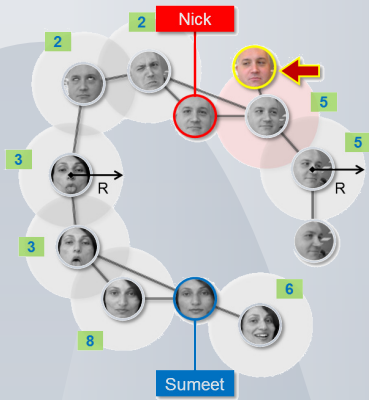
Online SSL with Graphs: Graph Quantization



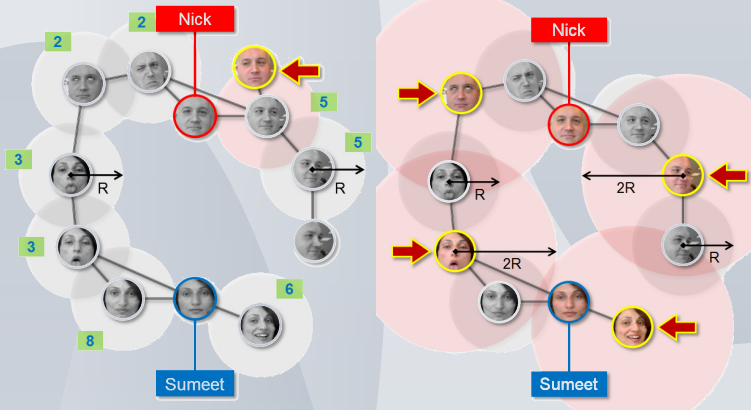
Online SSL with Graphs: Graph Quantization



Online SSL with Graphs: Graph Quantization



Online SSL with Graphs: Graph Quantization



Online SSL with Graphs: Graph Quantization

Online k -centers

1: an unlabeled $\mathbf{x}_t$, a set of centroids C_{t-1} , multiplicities $\mathbf{v}_{t-1}$

Online SSL with Graphs: Graph Quantization

Online k -centers

- 1: an unlabeled $\mathbf{x}_t$, a set of centroids C_{t-1} , multiplicities $\mathbf{v}_{t-1}$
- 2: **if** ($|C_{t-1}| = k + 1$) **then**
- 3: $R \leftarrow mR$

Online SSL with Graphs: Graph Quantization

Online k -centers

- 1: an unlabeled $\mathbf{x}_t$, a set of centroids C_{t-1} , multiplicities $\mathbf{v}_{t-1}$
- 2: **if** ($|C_{t-1}| = k + 1$) **then**
- 3: $R \leftarrow mR$
- 4: greedily repartition C_{t-1} into C_t such that:
- 5: no two vertices in C_t are closer than R
- 6: for any $\mathbf{c}_i \in C_{t-1}$ exists $\mathbf{c}_j \in C_t$ such that $d(\mathbf{c}_i, \mathbf{c}_j) < R$

Online SSL with Graphs: Graph Quantization

Online k -centers

- 1: an unlabeled $\mathbf{x}_t$, a set of centroids C_{t-1} , multiplicities $\mathbf{v}_{t-1}$
- 2: **if** ($|C_{t-1}| = k + 1$) **then**
- 3: $R \leftarrow mR$
- 4: greedily repartition C_{t-1} into C_t such that:
- 5: no two vertices in C_t are closer than R
- 6: for any $\mathbf{c}_i \in C_{t-1}$ exists $\mathbf{c}_j \in C_t$ such that $d(\mathbf{c}_i, \mathbf{c}_j) < R$
- 7: update $\mathbf{v}_t$ to reflect the new partitioning
- 8: **else**
- 9: $C_t \leftarrow C_{t-1}$
- 10: $\mathbf{v}_t \leftarrow \mathbf{v}_{t-1}$

Online SSL with Graphs: Graph Quantization

Online k -centers

- 1: an unlabeled $\mathbf{x}_t$, a set of centroids C_{t-1} , multiplicities $\mathbf{v}_{t-1}$
- 2: **if** ($|C_{t-1}| = k + 1$) **then**
- 3: $R \leftarrow mR$
- 4: greedily repartition C_{t-1} into C_t such that:
- 5: no two vertices in C_t are closer than R
- 6: for any $\mathbf{c}_i \in C_{t-1}$ exists $\mathbf{c}_j \in C_t$ such that $d(\mathbf{c}_i, \mathbf{c}_j) < R$
- 7: update $\mathbf{v}_t$ to reflect the new partitioning
- 8: **else**
- 9: $C_t \leftarrow C_{t-1}$
- 10: $\mathbf{v}_t \leftarrow \mathbf{v}_{t-1}$
- 11: **end if**
- 12: **if** $\mathbf{x}_t$ is closer than R to any $\mathbf{c}_i \in C_t$ **then**
- 13: $\mathbf{v}_t(i) \leftarrow \mathbf{v}_t(i) + 1$
- 14: **else**

Online SSL with Graphs: Graph Quantization

Online k -centers

```
1: an unlabeled  $\mathbf{x}_t$ , a set of centroids  $C_{t-1}$ , multiplicities  $\mathbf{v}_{t-1}$ 
2: if ( $|C_{t-1}| = k + 1$ ) then
3:    $R \leftarrow mR$ 
4:   greedily repartition  $C_{t-1}$  into  $C_t$  such that:
5:     no two vertices in  $C_t$  are closer than  $R$ 
6:     for any  $\mathbf{c}_i \in C_{t-1}$  exists  $\mathbf{c}_j \in C_t$  such that  $d(\mathbf{c}_i, \mathbf{c}_j) < R$ 
7:   update  $\mathbf{v}_t$  to reflect the new partitioning
8: else
9:    $C_t \leftarrow C_{t-1}$ 
10:   $\mathbf{v}_t \leftarrow \mathbf{v}_{t-1}$ 
11: end if
12: if  $\mathbf{x}_t$  is closer than  $R$  to any  $\mathbf{c}_i \in C_t$  then
13:    $\mathbf{v}_t(i) \leftarrow \mathbf{v}_t(i) + 1$ 
14: else
15:    $\mathbf{v}_t(|C_t| + 1) \leftarrow 1$ 
16: end if
```

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997** incremental

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

R

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R}$$

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2}$$

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2} + \dots = R \left(1 + \frac{1}{R} + \frac{1}{R^2} + \dots \right)$$

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997** **incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2} + \dots = R \left(1 + \frac{1}{R} + \frac{1}{R^2} + \dots \right) = \frac{RR}{R-1}$$

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997** incremental

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2} + \dots = R \left(1 + \frac{1}{R} + \frac{1}{R^2} + \dots \right) = \frac{RR}{R-1}$$

Guarantees

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2} + \dots = R \left(1 + \frac{1}{R} + \frac{1}{R^2} + \dots \right) = \frac{RR}{R-1}$$

Guarantees: 8-approximation algorithm.

Online SSL with Graphs: Graph Quantization

Doubling algorithm **charikar1997incremental**

To reduce growth of R , we use $R \leftarrow R \times R$, with $R \geq 1$

C_t is changing. How far can x be from some c ?

$$R + \frac{R}{R} + \frac{R}{R^2} + \dots = R \left(1 + \frac{1}{R} + \frac{1}{R^2} + \dots \right) = \frac{RR}{R-1}$$

Guarantees: 8-approximation algorithm.

Why not incremental k -means?

Online SSL with Graphs

Video examples

<http://www.bkveton.com/videos/Coffee.mp4>

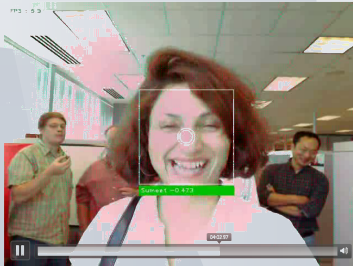
<http://www.bkveton.com/videos/Ad.mp4>

<https://misovalko.github.io/publications/kveton2009nipsdemo.adaptation.mov>

<https://misovalko.github.io/publications/kveton2009nipsdemo.officespace.mov>

<https://misovalko.github.io/publications/press-intel-2015.mp4>

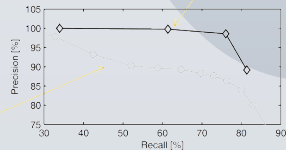
SSL with Graphs: Some experimental results



- 8 people classification
- Making funny faces
- 4 faces/person are labeled

Nearest Neighbor

Our method



Michal Valko

`michal.valko@inria.fr`

Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>